LotScope PHL — Product Requirements Document (Full Vision)

> **Working title:** *LotScope PHL* (placeholder — rebrand at will) **Document type:** Full-feature PRD (ideal-state, **not** MVP-scoped) **Version:** 0.1 · **Date:** 2026-06-27 · **Owner:** Turaboy · **Status:** Draft for refinement **How to read this:** This is the complete platform vision. Every feature carries a tier tag — **Core**, **Differentiator**, **Growth**, **Adjacent** — so you can extract an MVP later by filtering, without losing the big picture. Nothing here is pre-cut.

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1. Vision & Thesis

One line: The single screen that tells anyone — investor, builder, agent, or owner — *what can be built on any Philadelphia vacant lot, what it's worth, and what's wrong with it*, with the reasoning shown and the sources cited.

The wedge. Philadelphia already publishes the richest property data of any major US city — parcels, assessments, real recorded sale prices, zoning, L&I, demolitions, tax delinquency, the Land Bank, and the city's own vacancy model. The raw data exists; nobody has turned it into an *answer*. National tools (PRYCD, LandApp, Acres, Land Portal) are a mile wide and an inch deep, built for rural/exurban raw land, and they cannot reason about Philadelphia's zoning code, its by-right envelopes, its liens and sheriff sales, or its Land Bank. LotScope wins by being Philadelphia-deep: a reasoning layer on top of the city's full data stack, plus real comps (which Texas-style non-disclosure markets can't offer), plus an owner-side revenue stream the generic tools ignore.

Why this can be monetized in Philadelphia specifically. A hot, active infill market with tens of thousands of vacant parcels; a defined local audience of rowhouse developers, flippers, wholesalers, agents, and CDCs who pay for deal flow and due diligence; a large pool of tax-delinquent and over-assessed parcels that supports an owner-side appeal business; and free, high-quality city data that keeps cost of goods low and margins high.

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2. Problem Statement

Evaluating a Philadelphia vacant lot today means manually cross-referencing at least seven systems — OPA assessments, DOR recorded transfers, Zoning Base Districts, L&I violations/permits/demolitions, the tax-delinquency dataset, the Land Bank/city-owned inventory, and FEMA flood — then interpreting Title 14 of the zoning code to figure out what can be built by-right. That is hours of work per parcel, it requires expertise most buyers don't have, and a single missed flag (a tangled title, a six-figure tax lien, a lot too narrow to build on by-right) turns into an expensive mistake. There is no tool that does this end-to-end for Philadelphia and returns a plain answer.

Jobs to be done:

- *Find* the vacant lots worth my time, filtered by what matters to me.
- *Judge* whether a specific lot is a good buy and at what price.
- *Understand* what I can legally build on it, in plain English.
- *Avoid* the deal-killers (liens, title, zoning, flood, demolition).
- *Act* — reach the owner, model the deal, make an offer, manage my pipeline.
- (Owner side) *Know* what my lot is worth, whether I'm over-assessed, and how to sell or appeal.

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3. Market & Opportunity (Philadelphia)

- **Active infill market.** Zoning-permit filings have intensified over the past decade; developers are bullish on previously overlooked vacant parcels, pushing up land value.
- **Large addressable inventory.** The city's Vacant Property Indicators model flags tens of thousands of likely-vacant lots citywide — your core inventory, handed to you for free.
- **Real comps.** Unlike non-disclosure states, Philadelphia's DOR publishes recorded sale prices (with realty transfer tax) back to 1999 — making honest valuation possible.
- **Distress supply for the owner-side business.** A routinely-updated tax-delinquency dataset plus sheriff-sale activity supplies both motivated-seller leads and tax-appeal candidates.
- **Low cost of goods.** The bulk of the data is free, open, and API-accessible (Carto + ArcGIS Hub), so gross margin is structurally high.

TAM framing (Philadelphia). Paying segments: small/infill developers and builders, flippers/wholesalers, land-focused agents and brokerages, buy-and-hold land investors, CDCs/nonprofits, and lot owners. The single-market ceiling is intentional — go deep, dominate, then port the same engine to other rich-open-data cities (NYC, Chicago, DC, LA) as an expansion play.

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4. Personas

Persona	Goal	Key pains	Willingness to pay	Notes
Infill developer / small builder (primary)	Buy lots, build rowhouses/duplexes/triplexes for profit	By-right envelope, ARV, profit math, zoning/permit risk	High — pays for deal flow + DD + feasibility	The core payer; deepest feature needs
Land flipper / wholesaler	Buy cheap, assign/resell at volume	Sourcing, owner contact, pricing, off-market lists, mail	High — pays for lead gen + outreach	Volume user; outreach-heavy
Buy-and-hold land investor	Acquire/hold lots for appreciation, side yards	Valuation, lien/tax checks, portfolio tracking	Medium	Long-tail, sticky
Individual / owner-occupant buyer	Buy adjacent lot, build a home, garden	"Can I do X here," value, process	Low subscription; per-report or freemium	High volume, low LTV; transactional monetization
Real estate agent / broker	Serve land buyers/sellers, farm a neighborhood	Client-ready reports, listing intel, comps	Medium-high (pro/white-label)	White-label channel
CDC / nonprofit / community group	Neighborhood development, anti-blight, land banking	Vacancy, ownership, disposition data	Low (civic tier)	Credibility + partnership channel
Lot owner / over-assessed owner	Sell, appeal taxes, understand value	Don't know value, over-paying taxes, tangled title	Contingency or flat fee	Powers the Adjacent revenue lines

5. Feature Catalog (Full Set)

Tier legend: **Core** = the spine; **Differentiator** = what wins; **Growth** = scale/monetization;

Adjacent = separate but synergistic bets.

5.1 Map, Search & Discovery

Feature	Description & user value	Tier
Interactive vector parcel map	Fast pan/zoom map of every Philadelphia parcel; the front door.	Core
Vacant-lot layer & color-by	Toggle vacant lots; color by zoning, value, buildability, vacancy confidence, or opportunity score.	Core
Multi-criteria filters	Filter by size, dimensions, zoning, est. value, by-right units, owner type, vacancy, flood, lien status, neighborhood, council district, etc.	Core
Draw / radius / neighborhood search	Search within a drawn polygon, radius, or named neighborhood/RCO.	Core
Universal search	Address, parcel number, or owner search (Elasticsearch-backed).	Core
Saved searches + alerts	Save criteria; get email/push alerts when new matching lots appear.	Differentiator
Map data layers	Zoning, flood, historic districts, opportunity zones, council districts, neighborhoods, RCO boundaries, transit, schools, crime.	Core
Imagery	Aerial/satellite + Street View + historic aerials (GeoHistory) for change-over-time.	Core
Heat overlays	Permit activity, sales velocity, investor demand, vacancy density.	Growth
Parcel compare	Side-by-side comparison of multiple lots.	Differentiator

5.2 Parcel Dossier / Intelligence

Feature	Description & user value	Tier
Full parcel profile	Address, parcel #, owner, mailing address, dimensions, area, zoning, land use, assessment & sale history in one panel.	Core
Ownership detail + owner portfolio	Current owner plus all other parcels they own (assemblage / motivated-seller signal).	Differentiator

Feature	Description & user value	Tier
Assessment & tax history	OPA market/land/building values over time.	Core
Transfer history	DOR recorded sales/deeds/mortgages/sheriff deeds with prices.	Core
AI plain-language summary	One-paragraph plain-English read of the parcel.	Differentiator
Lot-condition from imagery (CV)	Computer vision on aerials: cleared / overgrown / debris / has structure.	Differentiator
Adjacent parcels & assemblage	Flag neighboring lots and consolidation opportunities.	Growth
Utilities proximity	Water/sewer/electric/gas availability indicators.	Differentiator
Neighborhood context	Median values, recent sales, permits, demographics around the lot.	Growth

5.3 Developability / Buildability Engine *(Philadelphia core brain)*

Feature	Description & user value	Tier
Zoning resolution + overlays	Resolve each lot's base district (RSA/RM/CMX/etc.) and overlays (historic, NCO, /CTR).	Core
By-right use determination	What's allowed without a variance: single-/two-/multi-family, commercial, mixed-use.	Differentiator
By-right dimensional analysis	Compare lot area/width to district minimums → conforming vs. non-conforming.	Differentiator
Buildable envelope	Max by-right units, height, occupied/open area, setbacks, required parking.	Differentiator
Variance / ZBA likelihood	Flag when a desired use/size needs a variance or special exception.	Differentiator
Build-scenario modeling	"1 unit vs 3 units" feasibility with by-right vs. variance paths.	Differentiator
Subdivision / consolidation potential	Whether the lot can be split or merged for a better outcome.	Growth

Feature	Description & user value	Tier
Overlay & RCO impacts	Historic review, neighborhood conservation, required community notification.	Growth
Buildable-area calc	Net buildable area after flood, setbacks, easements.	Differentiator
Buildability score + reasons	A single score with the make-or-break reasons shown.	Core

> Dimensional standards are populated from **Title 14 of the Philadelphia Code** into a versioned lookup table — never hardcoded blindly, since the code is amendable.

5.4 Valuation & Deal Analysis

Feature	Description & user value	Tier
Land value estimate (AVM)	Comp-based estimate with a confidence score.	Differentiator
Comparable sales finder	Real recorded (RTT) comps, filterable, mapped.	Core
ARV / finished-product estimate	Value of the completed build for a given scenario.	Differentiator
Deal score / underpriced flag	Flag lots priced below the model — "deal radar."	Differentiator
Feasibility / profit calculator	Acquisition + construction + soft costs + sale → margin / ROI.	Differentiator
Max allowable offer (MAO)	Back-solve the offer that hits a target margin.	Growth
Hold / build-to-rent modeling	Rent and hold scenarios for buy-and-hold users.	Growth
Assessment-vs-market gap	Over/under-assessment signal (feeds the tax-appeal product).	Differentiator
Scenario comparison	Compare multiple deal structures on one parcel.	Growth

5.5 Risk & Due Diligence

Feature	Description & user value	Tier
Tax delinquency / lien check	Amount, years owed, sheriff-sale exposure.	Core
City / Land Bank / PHDC ownership	Flag publicly-owned lots and disposition status.	Core

Feature	Description & user value	Tier
L&I status	Violations, permits, demolitions, unsafe/imminently-dangerous flags.	Core
Title red flags	Multiple/ deceased owners, tangled title, conservatorship signals.	Differentiator
Flood & environmental	FEMA flood zone, contaminated sites / brownfields proximity.	Core
Access / easements	Easements, rights-of-way, landlocked detection.	Differentiator
Conservatorship / Act 135 candidates	Distressed-property legal pathways.	Adjacent
Sheriff / tax sale calendar	Upcoming sales and parcel status.	Growth
Auto due-diligence checklist	Per-parcel checklist auto-populated from the data.	Differentiator
Risk score + flag list	Consolidated risk read with every flag itemized.	Core

5.6 AI Assistant — "Ask the Parcel" *(flagship differentiator)*

Feature	Description & user value	Tier
Grounded natural-language Q&A	"Can I build a triplex? Do I need a variance? Is this in the Land Bank?" — answered from zoning code + parcel data, **with citations**.	Differentiator
Deed / restriction extraction	RAG/OCR over recorded deeds & plats to surface restrictions ("no multi-family," easements).	Differentiator
Zoning code interpretation	Per-parcel reading of Title 14 with the specific section cited.	Differentiator
Process guidance	What permits/steps are needed to build X (ZBA, RCO, L&I).	Differentiator
Document summarization	Summarize deeds, plats, zoning notices in plain English.	Growth
Cross-parcel Q&A	Ask questions across a saved set ("which of these allows 3 units?").	Growth
AI report drafting	Generate narrative sections of dossiers and investor reports.	Growth

> **Guardrail:** every AI answer is framed as informational (not legal advice), shows confidence, and cites the source section/document.

5.7 Sourcing & Outreach

Feature	Description & user value	Tier
Off-market lead lists	Build lists like "vacant + delinquent + absentee owner."	Growth
Owner skip-trace	Append phone/email via provider.	Growth
Direct mail campaigns	Postcards/letters with templates and response tracking.	Growth
SMS / email sequences	Outreach automation to owners.	Growth
Absentee / out-of-state targeting	Filter for owners likely to sell.	Growth
Motivated-seller scoring	Score by delinquency + vacancy + absentee + ownership age.	Differentiator
Land Bank acquisition workflow	Eligibility, side-yard / EOI, application tracking.	Adjacent
Sheriff / tax sale lead tracking	Track and act on auction pipelines.	Growth
Lead status & response tracking	Manage outreach outcomes.	Growth

5.8 Market Intelligence & Analytics

Feature	Description & user value	Tier
Neighborhood dashboards	Values, sales velocity, permits, vacancy trends per area.	Growth
Price & rent trend charts	By neighborhood / zip / zoning.	Growth
Development-activity heatmaps	Where permits are filed and developers are buying.	Growth
Comp trend analytics	How land comps move over time.	Growth
Incentive overlays	Opportunity zones, tax abatements, special districts.	Growth

Feature	Description & user value	Tier
Investor-activity intel	Who is buying, and where.	Growth
Vacancy & blight trend tracking	Track change at neighborhood level.	Growth
Custom market reports	Generate area reports on demand.	Growth

5.9 Portfolio, Pipeline & Collaboration

Feature	Description & user value	Tier
Watchlists	Save parcels with tags and notes.	Core
Deal pipeline	Kanban: lead → DD → offer → under contract → closed.	Growth
CRM-lite	Track contacts, owners, sellers.	Growth
Per-deal document management	Store docs against each deal (S3).	Growth
Notes, tasks, reminders	Lightweight productivity per parcel/deal.	Growth
Team collaboration	Shared lists, comments, assignment.	Growth
Price/status monitoring	Auto-alerts on changes to saved lots.	Differentiator
Activity history / audit	Track what changed and when.	Growth

5.10 Reports & Export

Feature	Description & user value	Tier
One-click PDF dossier	Branded full parcel report.	Differentiator
Investor / lender / partner report	Decision-ready package to share.	Growth
Comp report	Standalone valuation comps.	Growth
Feasibility / pro-forma report	The deal math, exportable.	Growth
Shareable web links	Send a live parcel view.	Growth
CSV / Excel export	Bulk data out.	Growth
White-label / co-branded reports	Agent/brokerage branding.	Growth

Feature	Description & user value	Tier
Scheduled / auto reports	Recurring deliverables.	Growth

5.11 Workflow & Transaction Tools

Feature	Description & user value	Tier
Offer / LOI generation	Auto-draft offers from parcel data.	Growth
Purchase agreement templates	PA/Philadelphia-appropriate templates.	Growth
E-signature integration	Sign without leaving the platform.	Growth
DD checklist & tracking	Run due diligence to close.	Growth
Title / closing handoff	Integrate or hand off to a title company.	Growth
Document vault	Secure per-transaction storage.	Growth
Transaction timeline	Track milestones to closing.	Growth

5.12 Owner-Side & Tax Appeal *(Adjacent — same data, different customer)*

Feature	Description & user value	Tier
Over-assessment scoring	Rank parcels assessed above model/comps.	Adjacent
Tax-appeal eligibility + savings estimate	Show likely savings before filing.	Adjacent
Appeal assistance / filing support	First Level Review / BRT appeal workflow.	Adjacent
"Sell my vacant lot"	Owner valuation + buyer matching / lead gen.	Adjacent
Owner dashboard	My parcels, taxes, value, inbound offers.	Adjacent
Owner outreach (compliance/sell)	Reach owners of nuisance/vacant lots.	Adjacent

5.13 Marketplace & Lead Gen *(Growth — network effects)*

Feature	Description & user value	Tier
Off-market listing/marketplace	Owners list lots; buyers browse.	Growth

Feature	Description & user value	Tier
Buyer–seller matching	Match motivated owners to active buyers.	Growth
Lead marketplace	Sell qualified motivated-owner leads (referral/take-rate).	Adjacent
Featured listings	Paid promotion.	Growth
Agent directory	Connect users to land-savvy agents.	Growth

5.14 Accounts, Teams, Billing & Admin

Feature	Description & user value	Tier
Auth & roles	Email/SSO, role-based permissions.	Core
Subscriptions & billing	Tiered plans (Stripe), proration, invoices.	Core
Usage metering / credits	Meter skip-trace, mail, report credits.	Growth
Team seats & management	Add/manage seats and roles.	Growth
Tier limits	Enforce feature/usage caps per plan.	Core
Admin console	Data status, user/content management.	Core
API keys & developer portal	For the API/data tier.	Growth
White-label config	Brokerage/agent branding.	Growth

5.15 Data Platform & Infrastructure *(the moat, treated as product)*

Feature	Description & user value	Tier
Ingestion pipelines	Scheduled ETL from Carto SQL, ArcGIS Hub, FEMA, Regrid, deeds.	Core
Freshness monitoring + transparency	SLAs and visible "last updated" per dataset.	Core
Parcel-key reconciliation	Resolve OPA ↔ PWD ↔ RTT identity mismatches.	Core
Verdict computation engine	Batch + on-demand computation of the parcel verdict.	Core
Vector tile serving	Self-hosted tiles (Martin/pg_tileserv) for cost control.	Core

Feature	Description & user value	Tier
Search index	Elasticsearch for address/owner/parcel search.	Core
Document store + OCR	S3 + OCR pipeline for deeds/plats feeding RAG.	Differentiator
Data-quality & confidence scoring	Per-field quality and confidence surfaced to users.	Differentiator
Verdict versioning / audit	Reproducible, auditable verdicts.	Growth

5.16 Notifications & Engagement

Feature	Description & user value	Tier
New-match alerts	Saved-search hits.	Differentiator
Watchlist change alerts	Price/status changes on saved lots.	Differentiator
Disposition alerts	New sheriff-sale / Land Bank lots in target area.	Growth
Permit-nearby alerts	Development signal near saved lots.	Growth
Digest emails	Weekly opportunity digest.	Growth
In-app + push	Real-time engagement (mobile).	Growth

5.17 Mobile & Field Tools

Feature	Description & user value	Tier
Mobile app / PWA	LotScope on the go.	Growth
"Lots near me"	Geolocated discovery.	Growth
Drive-for-dollars	Log lots from the field.	Growth
Geotagged photos	Attach field photos to parcels.	Growth
Offline saved parcels	Access without signal.	Growth
On-the-go verdict	Quick verdict in the field.	Growth

5.18 Trust, Compliance & Legal *(cross-cutting)*

Feature	Description & user value	Tier
Source citations everywhere	Every data point and AI answer cites its source.	Core
Confidence indicators	Show certainty, never false precision.	Core
"Not legal/financial advice" framing	Clear disclaimers throughout.	Core
Data-license compliance	Honor city "as-is" terms, Regrid, provider licenses.	Core
Outreach compliance	TCPA / DNC / CAN-SPAM for mail/SMS/email.	Growth
Fair Housing compliance	Guardrails on targeting/marketing.	Growth
PII security & access control	Protect owner contact data.	Core
Audit trail	Record verdicts and actions.	Growth

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6. Flagship Spec — The "Parcel Verdict"

The product's soul is one screen. When a user clicks any vacant lot:

1. **Headline verdict** — a single buildability/opportunity score with a one-line plain-English summary.
2. **What you can build** — by-right use(s), max units, height, and whether a variance is needed; a 1-line "buildable envelope" plus alternate scenarios.
3. **What it's worth** — a confidence-scored value range, the top comps (mapped), and the assessment-vs-market gap.
4. **What's wrong with it** — itemized red flags: liens/delinquency, title issues, L&I/demolition, Land Bank ownership, flood, access.
5. **Ask the parcel** — an inline Q&A box for plain-English questions, answered with citations.
6. **Act** — save to watchlist, generate PDF, model the deal, or reach the owner.

Every element shows its source and last-updated date. The verdict is the free hook in limited form; the full verdict + Q&A + reports sit behind the paywall.

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7. AI / ML Components

- **RAG Q&A engine** — retrieval over (a) Title 14 zoning code text, (b) the parcel's resolved district + dimensions, (c) recorded deeds/plats; generation grounded and

cited (Claude API + vector store).

- **Document understanding** — OCR + extraction of restrictions, easements, and conditions from recorded documents.
- **Valuation model** — comp-based AVM trained on RTT recorded sales, with confidence/coefficient-of-variation handling for thin sub-markets.
- **Vacancy refinement** — improve on the city's Vacant Property Indicators with imagery and assessment signals.
- **Lot-condition CV** — classify aerial imagery (cleared / overgrown / debris / structure present).
- **Motivated-seller / over-assessment scoring** — ranking models over delinquency, vacancy, absentee, ownership tenure, and assessment-vs-comp gap.

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8. Data Sources & Data Model (summary)

Tabular via Carto SQL (`https://phl.carto.com/api/v2/sql`): `opa_properties_public` (assessments/owner/value), `pwd_parcels` (parcel polygons), `rtt_summary` (recorded transfers/comps), real estate tax delinquencies, L&I violations/permits/demolitions. **GIS via ArcGIS Hub (`data-phl.opendata.arcgis.com`):** Vacant Indicators – Land, Zoning Base Districts, building footprints, land use, FEMA flood, historic districts, council districts, neighborhoods/RCOs. **Publicly-owned / Land Bank inventory:** city-owned property dataset (locate current source). **Recorded documents:** DOR records system for deed/plat images (RAG layer). **Third-party (optional):** Regrid (standardized ownership + mailing for skip-trace; portability beyond Philadelphia), skip-trace provider, imagery provider.

Data-model spine: a master `parcel` table keyed on OPA `parcel_number`, with related `zoning_district`/`zoning_standard`, `sale`, `tax_status`, `li_event`, and a computed `parcel_verdict` materialized view. Store geometry in EPSG:2272 (analysis) and 4326 (tiles). Known gotchas: Carto times out on full-table dumps (chunk by parcel ranges); OPA/PWD/RTT join keys need reconciliation; case-sensitive SQL (use `ILIKE`).

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9. Monetization & Pricing

Multiple revenue lines; the generic competitors floor (PRYCD ≈ $25/mo, LandApp ≈ $6/mo) means **don't compete on raw data price** — charge a premium justified by Philadelphia depth, the answer layer, and workflow, and capture revenue the generic tools can't (pro/agency, owner-side, marketplace).

Stream	Model	Indicative price	Target
Free	Freemium hook	$0	Vacant-lot map, basic parcel info, a few verdicts/mo
Pro	Subscription	~$49–99/mo	Unlimited verdicts, comps, AI Q&A, watchlist, alerts, reports — individual investors/flippers
Team / Agency	Subscription + seats	~$199–399/mo	Pipeline/CRM, outreach, advanced analytics, white-label — developers, brokerages
Enterprise / Data / API	Custom	Custom	Data feed, API, integrations — institutions, lenders, CDCs, the city ecosystem
À la carte	Transactional credits	Report ~$10–25; skip-trace & mail credits	Occasional/retail buyers; outreach-heavy users
Owner-side: tax appeal	Contingency or flat fee	% of first-year savings, or flat	Over-assessed lot owners
Owner-side: sell my lot	Lead gen / referral	Per-lead or take-rate	Motivated owners → investors/agents
Marketplace	Listing fee / take-rate	% or flat	Facilitated transactions (later)

10. Go-to-Market (Philadelphia)

- **Top of funnel:** the free, genuinely-useful vacant-lot map — the best free Philadelphia lot explorer in existence — plus Philadelphia-specific content/SEO ("can I build a duplex on [zoning] in Philly?").
- **Community:** local REIAs, Philadelphia developer/investor groups, wholesaler networks, and meetups; dogfood and credibility via your own lot.
- **Partnerships:** title companies, land-savvy agents/brokerages (white-label), CDCs and the Land Bank ecosystem (civic tier + credibility), and lenders.
- **Owner-side acquisition:** direct outreach to delinquent/over-assessed owners (the tax-appeal funnel doubles as a seller funnel).
- **Expansion later:** replicate to other rich-open-data cities once Philadelphia is dominated.

11. Competitive Landscape & Differentiation

Competitor	What they do	Why LotScope wins in Philadelphia
PRYCD	Land valuation, skip-trace, mail (rural-focused, US-wide, ~$25/mo)	No Philadelphia zoning reasoning, no by-right answer, weak in urban infill
LandApp / LandGate	40+ national layers, energy-focused (~$6/mo)	Energy/rural lens; no Title 14 interpretation, liens, Land Bank, or local comps
Acres.com	Due-diligence panel + portfolio	National breadth, not Philadelphia depth or owner-side
Pebble	Land-investor CRM	CRM only; no intelligence/answer layer
Land Portal	Map + AI vacant filter + filters	Rural raw-land orientation; no urban-infill reasoning
Zoneomics	National zoning data	Data layer, not a Philadelphia answer/workflow product
PropStream	General REI data (160M props)	Houses-first, generic; no land/infill or by-right modeling
Land.com / Zillow	Listings	Listings only; no off-market, no analysis, no DD

Moat: Philadelphia data depth + the reasoning/answer layer + real recorded comps + integrated risk (liens/title/L&I/Land Bank) + owner-side revenue — a combination none of the above has in this market.

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12. Non-Functional Requirements

- **Performance:** sub-second map pan/zoom; vector tiles served from own infra; verdict render < ~2s.
- **Freshness:** scheduled ingestion with visible "last updated"; daily for L&I/demolitions, regular for assessments/transfers/delinquency.
- **Accuracy & honesty:** confidence scores everywhere; no false precision; show the data and let users verify.
- **Scalability:** handle ~580k citywide parcels and tens of thousands of vacant lots; batch verdict recompute.

- **Security & privacy:** protect owner PII and skip-trace data; role-based access; encryption at rest/in transit.
- **Reliability:** graceful handling of upstream (Carto/ArcGIS) outages; cached layers.
- **Accessibility & mobile:** responsive/PWA; WCAG-minded UI.

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13. Legal, Compliance & Trust

- **AI = informational, not legal advice.** Cite the code section/document, show confidence, recommend confirming with L&I/zoning/an attorney. Get counsel on UPL exposure and disclaimers/ToS.
- **Data licensing.** City data is "as-is / hold-harmless"; honor those terms and Regrid/provider licenses; disclaim accordingly.
- **Outreach compliance.** TCPA/DNC for SMS/calls, CAN-SPAM for email, mail-list hygiene.
- **Fair Housing.** Guardrails on any targeting/marketing copy.
- **Auditability.** Reproducible verdicts and an action audit trail.

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14. Technical Architecture (high-level)

- **Data core:** PostGIS (parcels, layers, verdicts), Elasticsearch (search), S3 (documents/imagery), vector store (RAG).
- **Backend:** Python / FastAPI; verdict engine (batch + on-demand); ingestion ETL (Airflow or scheduled jobs) from Carto SQL + ArcGIS Hub + FEMA + Regrid + DOR.
- **AI services:** Claude API for Q&A/summaries; OCR + extraction pipeline; valuation & scoring models.
- **Frontend:** Quasar + MapLibre GL JS; self-hosted tiles (Martin/pg_tileserv).
- **Integrations:** Stripe (billing), skip-trace provider, mail house, e-sign, title/closing, imagery.
- **Infra:** AWS, Docker.

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15. Success Metrics (KPIs)

- **Acquisition/activation:** signups; first verdict viewed; first parcel saved.
- **Engagement:** WAU/MAU; verdicts/user; searches/user; reports generated.
- **Conversion/retention:** free→paid rate; churn; ARPU; MRR/ARR.
- **Product quality:** data coverage % and freshness lag; verdict accuracy validated against subsequent permits/sales; AI answer correctness; NPS.
- **Owner-side:** appeals filed; savings delivered; sell-side leads converted.

16. Risks & Mitigations

Risk	Mitigation
Data freshness/maintenance burden	Lean on city APIs + Regrid; freshness monitoring; visible "last updated."
Accuracy & legal liability (UPL)	Cite sources, show confidence, disclaim, get counsel.
Competitive/price pressure & commoditization	Compete on Philadelphia depth + answers + workflow + owner-side, not raw data price.
Single-market TAM ceiling	Intentional; expansion path to other rich-open-data cities.
Distribution risk	Community-led GTM; free map hook; partnerships.
Upstream data-access changes (city)	Abstract ingestion; cache; Regrid as fallback for base parcels.
Third-party dependency (skip-trace, imagery)	Provider-agnostic interfaces.

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17. Suggested Phasing (for later MVP extraction)

Full set above is the vision; a natural build order:

- **Phase 1 — Foundation (Core):** map + parcel dossier + developability engine + risk flags + basic valuation + the Parcel Verdict + free tier.
- **Phase 2 — Differentiation & monetization (Differentiator):** AI "Ask the Parcel," full comps/AVM, deal scoring, reports, watchlist/alerts, Pro billing.
- **Phase 3 — Pros & scale (Growth):** pipeline/CRM, outreach/skip-trace, team/agency tier, white-label, market analytics, mobile/field.
- **Phase 4 — Adjacent & network (Adjacent/Growth):** owner-side tax appeal, sell-my-lot lead gen, data/API, marketplace, multi-market expansion.

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18. Open Decisions

1. **Lead persona** — infill developer vs flipper vs owner-side first?
2. **Buyer-side vs owner-side sequencing** — which revenue line proves out first?
3. **Build vs license** non-city data (Regrid now or later)?

4. **Free-tier generosity** — how much verdict to give away for the hook?
5. **AI advice posture** — how conservative on the legal-interpretation feature?
6. **Partnership strategy** — city/CDC/title/agent — which first?
7. **Expansion timing & target cities.**

Appendix A — Glossary (Philadelphia)

- **OPA** — Office of Property Assessment (assessments; `opa_properties_public`).
- **RTT** — Realty Transfer Tax / DOR recorded transfers (`rtt_summary`; the comp source).
- **L&I** — Licenses & Inspections (violations, permits, demolitions).
- **PWD parcels** — authoritative parcel polygons.
- **Vacant Property Indicators** — the city's per-parcel vacancy model.
- **Land Bank / PHDC** — public land-holding/disposition entities.
- **RSA / RM / CMX** — Residential Single-family Attached / Residential Multi-family / Commercial Mixed-use zoning districts (Title 14).
- **ZBA** — Zoning Board of Adjustment (variances/special exceptions).
- **RCO** — Registered Community Organization (zoning notification).
- **Sheriff sale** — public auction of tax-/mortgage-foreclosed property.
- **Act 135** — Pennsylvania conservatorship statute for blighted property.

End of PRD v0.1 — refine and extract MVP from here.